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Evaluation of a Dynamic Model for Primary Infections Caused by *Plasmopara viticola* on Grapevine in Quebec

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Abstract

Downy mildew is major grape disease in several areas of the world. Recently, a dynamic model for primary infections of grapes by *Plasmopara viticola*, forecasting time of primary lesions emergence, was developed in Italy. The model simulates the development of predicted oospore cohorts during the primary infection period. The efficacy of this disease-cycle-based model was evaluated in eastern Canada by comparing the time of lesion emergence predicted by the model with field observations in 20 and 23 vineyards in 2008 and 2009, respectively. For each vineyard, one to 20 simulation runs were performed depending on the number of oospore cohorts expected to form, for a total of 545 simulations. The model evaluation was based on the true positive proportion (lesion emergence was predicted and observed) and the true negative proportion (lesion emergence was not predicted and not observed) which were 0.996, and 0.907, respectively. A total of 313 simulations resulted in no infection among which 284 corresponded to no lesion emergence. In only one situation, lesions were observed and not predicted by the model. On the contrary, in 29 simulations run, lesion emergence was predicted but not observed in the field. Further validation of this model is required, but the results of this study are encouraging and this model may be used to improve timing of fungicide sprays against *P. viticola*.

Introduction

In eastern Canada, as well as in several viticultural areas, downy mildew, caused by *Plasmopara viticola* (Berk et Curt.) Berlese et de Toni, represents one of the most important grapevine (*Vitis vinifera* L.) diseases. The warm and wet spring with abundant rainfall generally occurring in the spring favor disease development (13). The pathogen, native to North America, infects most species of *Vitis* with *vinifera* cultivars being highly susceptible and wild species relatively resistant. The pathogen attacks all aerial parts of the vines, causing indirect losses (leaf, shoot, and tendrils infections, Fig 1A) or direct losses (berry infections, Fig. 1B). In addition, the quality of wine produced from infected grapes is reduced (21). Grape downy mildew epidemics can progress rapidly and can cause economic losses up to 100%; hence, growers attempt to limit this threat by managing their fungicide program as though a disease risk is always present. However, increasing fungicide, fuel, and labor costs make routine use of fungicides costly. This strategy has several disadvantages, not only from an economic point of view but also with regard to the environmental impact of fungicides. In this context, the use of decision-support systems and of simulation models that predict “real risk” represents a valid alternative to calendar-based fungicide spray program.



Fig. 1. Sporulating colonies of *Plasmopara viticola* on young shoot and leaf (A) and berry (B).

Like most polycyclic diseases, epidemics of grape downy mildew entail a sequence of events initiated by the production and dispersal of initial inoculum, followed by primary infections, production and dispersal of secondary inoculum, and completed by survival of *P. viticola*. Recent molecular studies showed the preponderant effect of primary infections on the epidemic (8,9,24). Gobbin et al. (8,9) analyzed more than 10,000 isolates of *P. viticola* from 39 vineyards using microsatellite markers and found numerous infections originating from primary inoculum from May to August. Furthermore, they reported that genotypes identified once throughout the sampling period always constituted the dominant class (71% of all genotypes) regardless of the sampling date. Only one or two genotypes per epidemic developed into secondary cycles and generated a high number of progenies. In other words, epidemics of grape downy mildew are strongly influenced by primary inoculum and hence controlling them is crucial for adequate management of the disease.

The relationship of temperature and relative humidity to downy mildew development and *P. viticola* reproduction serve as the basis for most prediction systems. Grape downy mildew prediction models have been proposed for identifying the periods of high risk (i.e. conditions are favorable for disease development) and for scheduling fungicide applications (16,23). Some of these models are based on the simulation of primary infection development such as the POM (26), EPI (25), SIMPO (10), DMCAST (20), and UCSC (23) models. While other models predict the development of secondary infections through a simulation of one or more stages of *P. viticola* biological cycle (3,6,15,18,19).

Despite the availability of warning systems for management of grape downy mildew, Quebec's growers generally apply fungicides in a preventative manner starting soon after the green tip phenological stage or based on the 3-10 rule (2). This rule is based on the concurrent occurrence of: (i) air temperature equal to or greater than 10°C; (ii) vine shoots at least 10 cm long; (iii) a minimum of 10 mm of rainfall in the past 24 to 48 h. Considering the frequent spring to early summer occurrence of temperatures above 10°C and rainfall, this rule basically relies on the phenological criteria (vine shoots at least 10 cm long) and does not accounts for the development of the pathogen.

The objective of this study was to evaluate under the conditions of eastern Canada, the potential of a new dynamic model for predicting *P. viticola* development during the initial phase of the disease development (23) for predicting the emergence of the primary lesions.

Model Description

The model was previously described in detail (23). Briefly, it is a dynamic model developed following a mechanistic approach according to principles of systems analysis (22). The model simulates, with a time step of one hour, the entire process of downy mildew primary infection: oospore maturation and

germination, then zoospore ejection and dispersal, and finally infection establishment and disease symptom onset. This process was separated into different stages: the pathogen changes from one stage to another at different rates, depending on environmental conditions (Fig. 2). Pathogen stages are considered as state variables and their changes are regulated by rate variables; weather conditions influence rates acting as parameters or intermediate variables.

Hourly temperature (T), relative humidity (RH), and rainfall (R) from 1 January were used as input data in the model. A simulation was initiated by each rainfall wetting the leaf litter and triggering the oospore germination process. In this model, it is assumed that oospores are present. At this time, the oospores which have broken dormancy form a cohort which develops in a similar way. An assumption of the model is that the germination process is temperature dependent when humidity of the leaf litter is not a limiting factor. The model simulates all the primary infection cycle of *P. viticola* (23). After germination, sporangia present on the leaf litter release zoospore in the presence of water; otherwise sporangia can survive for a few days and then die, depending on temperature and relative humidity. Zoospores are splashed by rain droplets and aerosols to grape leaves, but if the litter surface dries before rainfall zoospores do not survive. Infection establishment is caused by deposited zoospores when wetness duration and the corresponding temperature are favorable. If the leaf litter surface dries the zoospores dry out and die. The length of incubation period is influenced by temperature and relative humidity. At the end of the incubation period lesions develop at the infection sites.

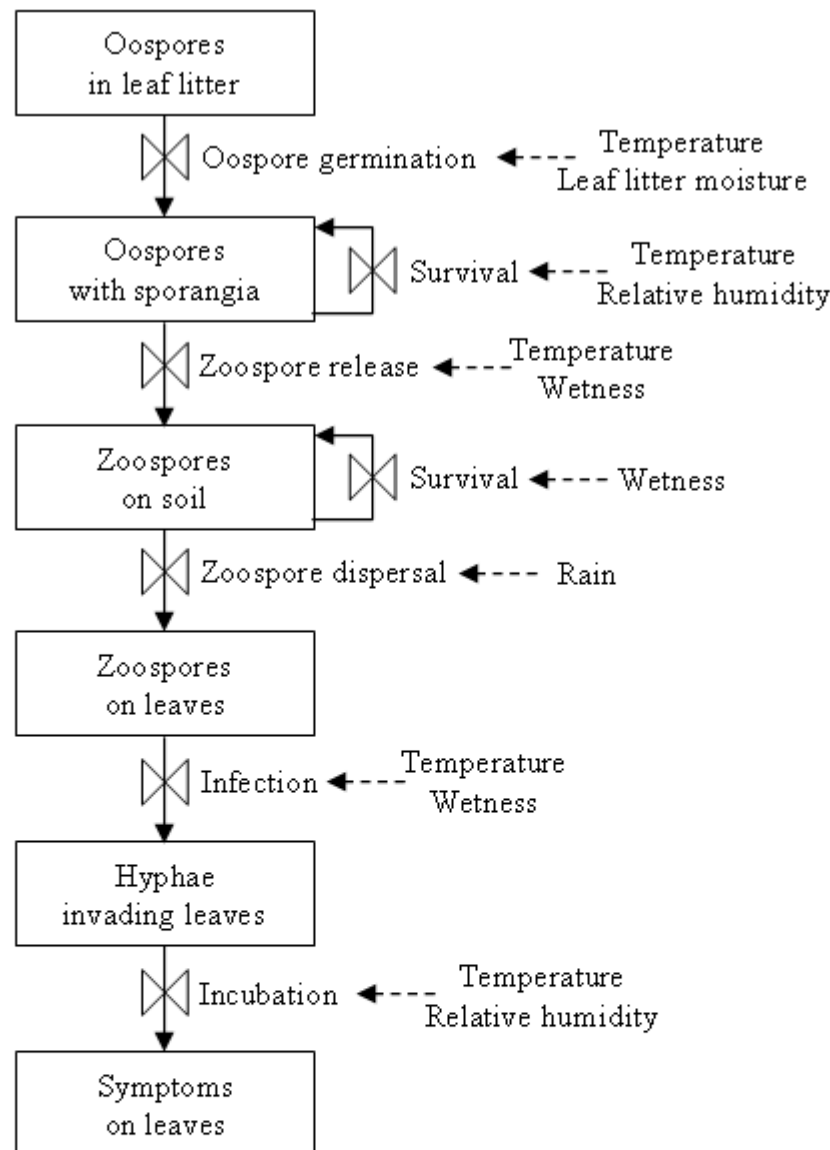


Fig. 2. Flow diagram of the model simulating the development of the *Plasmopara viticola* oospores from germination to disease appearance [redrawn from (18)]. The model considers the different stages of the primary infection cycle. Oospores form within affected leaf tissue and overwinter in leaf litter above or buried in the soil. During the winter, oospores break dormancy progressively and germinate under favorable environmental conditions. Germinating oospores form sporangia that survive for a few hours to a few days and release zoospores in the presence of water. Survival of the zoospores is strictly dependent on the presence of water. Viable zoospores on the soil surface are dispersed from the soil to grape leaves by rain splash. Once deposited on leaves, zoospores swim towards stomata, form a germ tube and penetrate the host. Infecting hyphae colonize the host tissue and the initial disease symptoms are noticed in the form of "oil spots". Legend: $\longrightarrow$ state variable; $\dashrightarrow$ flux and direction of states; $\square$ flux and direction of information; $\bowtie$ valve in a flux (rate).

Field Observations

The field observations used to evaluate the model were collected by government and private grape specialists and scouts in 20 and 23 vineyards in 2008 and 2009, respectively, located in the south of the province of Quebec, near the US border, and the north of the province (L'Île d'Orléans) (Fig. 3). Vineyards were selected based on availability of scouting data and the presence of susceptible cultivars. The vineyards selected for this evaluation are

representative of the different grapevine-growing areas, for soil type, varieties, training systems and cropping regimes. They also represent a range of dose of overwintering inoculum because of different fungicide spray programs, environmental conditions and geographical locations. Starting from bud burst, each vineyard was inspected at a 3 or 4 day-interval, to detect the emergence of the first lesions expressed as “oil spots” on leaves. At each assessment, all leaves on 50 to 100 vines, depending on vineyard size, were assessed for the presence of downy mildew lesions. Since the oil spots could have been appeared on each single day between the last negative and the first positive scouting, the actual symptoms onset was expressed as an “onset window” (Fig. 4).

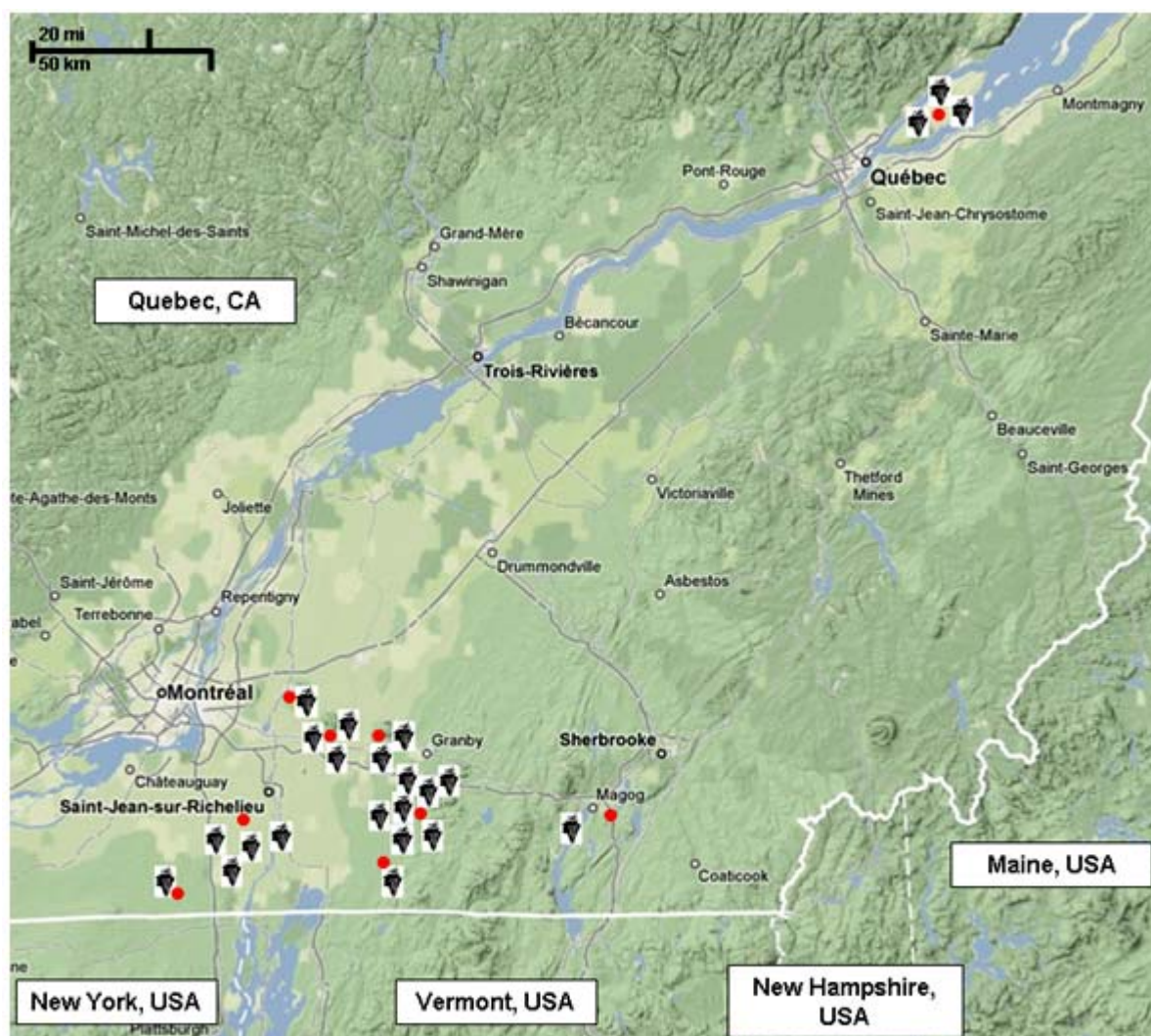


Fig. 3. Map of the province of Quebec representing locations of: weather stations (●) vineyards (🍷) used in this study.

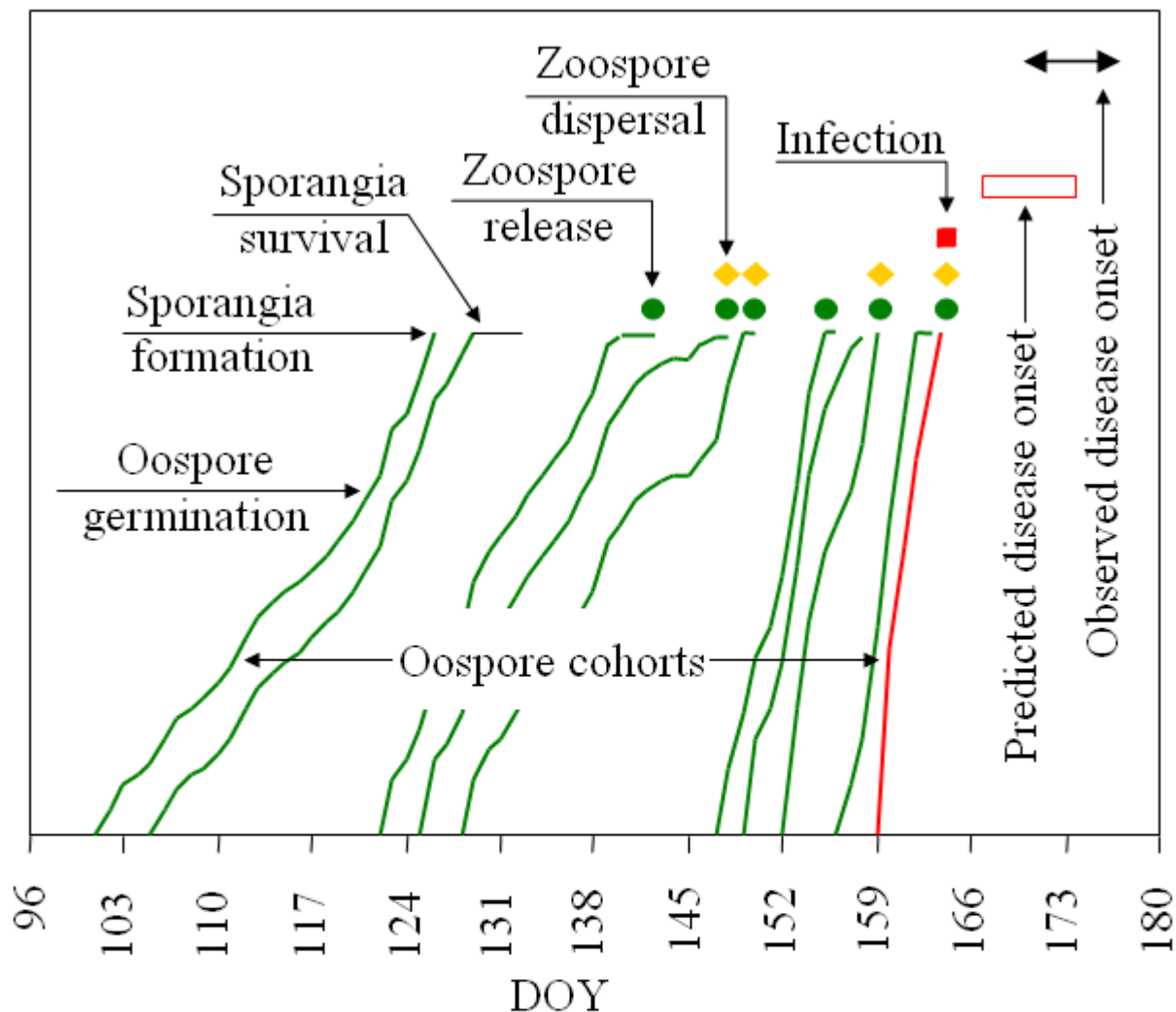


Fig. 4. Schematic representation of example outputs of a model simulating the development of the *Plasmopara viticola* oospores. The lines (— and —) show the germination course of oospore cohorts which did or did not cause infection, respectively. Other symbols represent: ■ zoospore release; ◆ zoospore dispersal; ■ infection; □ and $\longleftrightarrow$ estimated and actual periods of downy mildew onset, respectively.

Hourly temperature (°C), relative humidity (%) and rainfall (mm) were obtained from the nearest (not more than 15 km) automatic weather station operated by Environment Canada available. The model was used to simulate, for each year and each vineyard, the progress of downy mildew for each predicted oospore cohort from 1 January to the emergence of the lesions (23). For a small number of vineyards located near to each other data from the same weather station were used.

Methods for Model Evaluation

For each vineyard, one to 20 simulations were performed depending on the number of oospore cohorts expected to be formed (presence of rain), for a total of 545 simulations. All simulations were divided into two groups based on observed lesion emergence; the cases and the controls were defined as the simulation with lesion emergence observed (O+) or not observed (O-), respectively. Within each group, simulations were further divided into two groups based on positive (P+) and negative (P-) predictions. As a result all simulations were grouped into four groups: O+P+, O+P-, O-P+, and O-P- according to Caffi et al. (4) as shown in Table 1.

Table 1. Contingency table with number of true positive predictions, true negative predictions, false positive predictions and false negative predictions based on a total of 545 simulations of *Plasmopara viticola* lesion emergence.

	Lesion emergence predicted (P+)	Lesion emergence not predicted (P-)	Total
Lesion emergence observed (O+)	TPP ^v = 1.00 (231) ^e	FNP ^x = 0.00 (1)	232
Lesion emergence not observed (O-)	FPP ^w = 0.09 (29)	TNP ^y = 0.91 (284)	313
Total	260	285	545

^v The true positive proportion (TPP) was calculated by dividing the number of true positive predictions by the total number of cases (i.e. O+, P+, the disease was observed and predicted).

^w The false positive proportion (FPP or O-,P+) was calculated as 1 – TNP.

^x The false negative proportion (FNP or O+,P-) was calculated as 1 – TPP.

^y The true negative proportion (TNP) was calculated by dividing the number of true negatives predictions by the total number of controls (i.e. O-,P-, the disease was not observed and not predicted).

^z Numbers in parenthesis represent the number of simulations.

The sensitivity, defined as the true positive proportion (TPP) was calculated by dividing the number of true positive predictions by the total number of cases (i.e., O+, P+, the disease was observed and predicted). The specificity, defined as the true negative proportion (TNP) was calculated by dividing the number of true negatives predictions by the total number of controls (i.e. O-,P-, the disease was not observed and not predicted). The false positive proportion (FPP or O-,P+) was calculated as 1 – TNP and the false negative proportion (FNP or O+,P-) was calculated as 1 – TPP (12,17,27,28). Moreover, the probability that an oospore cohort, in a particular vineyard, results or not in the development of lesions was determined as $P(O+,P+)$ and $P(O-,P-)$, which are the probability that the lesions emerge when emergence is predicted and the probability that no lesions emerge when emergence is not predicted, respectively (17). These probabilities were compared with the corresponding prior probabilities, $P(O+)$ and $P(O-)$, respectively (28). The effectiveness of the model in predicting emergence of downy mildew lesion was assessed based on sensitivity, specificity, and overall accuracy (proportion of good predictions) (27,28).

Moreover, the most probable period of infection was calculated for all the locations going backward from the date of actual symptom onset according to Giosuè et al. (7) as shown in Fig. 5. The length of incubation period was calculated using two temperature dependent equations accounting for both low and high level of relative humidity. Within the time window obtained with this procedure, the probable infection date was determined as the day with the most favorable conditions of temperature and rainfall within the window (7). Then, the prediction of the first seasonal infection periods of *P. viticola* for each location were compared using three different predictors: (i) the UCSC model; (ii) the 3-10 rule (2); and (iii) the Kennelly's rule (11), which is a modified version of the 3-10 rule and is based on the concurrent occurrence of air temperature greater than 11°C, Eichorn and Lorenz growth stage 12 (5 to 6 leaves unfolded), and a minimum of 2.5 mm of rainfall.

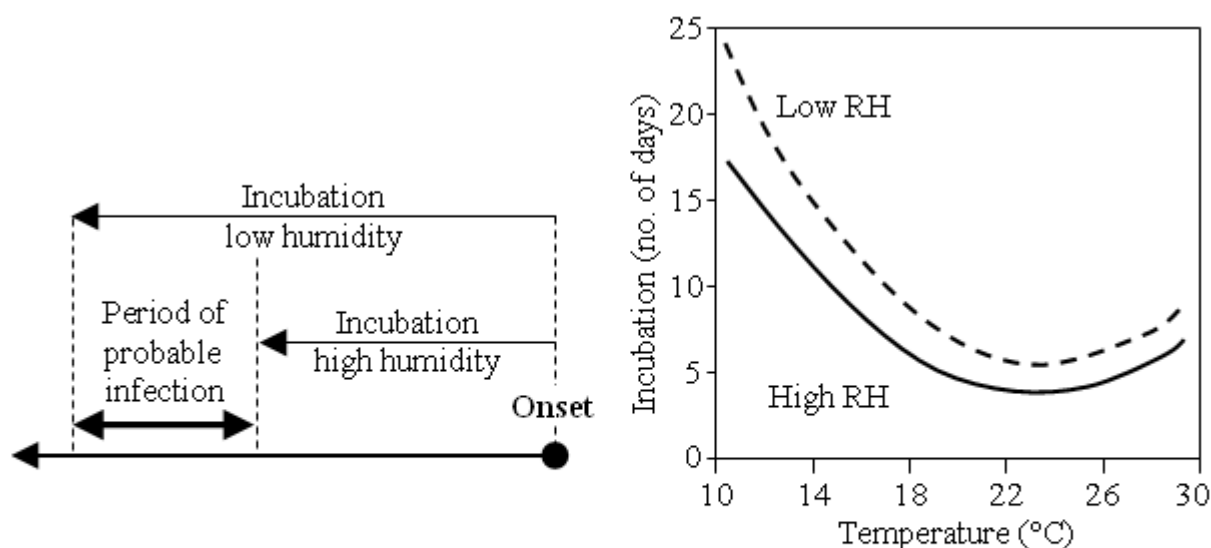


Fig. 5. Method for calculating the most probable date of infection going backward, from the date of actual onset reported during the field inspections, of one incubation period calculated according two temperature dependent equations for both high (—) and low (---) RH conditions (7).

Model Evaluation from Field Observations

The date of emergence of the first downy mildew lesions ranged between 8 and 29 June. On average, the first lesions were observed on 14 June (SD 6.2) in the south area, while there were observed more than a week later (23 June, SD 2.3) in the northern part of Quebec. In these periods, grapevines were between the growth stage “shoot 10 cm in length” and full flowering [stage 23 on the Eichhorn-Lorenz scale (5)].

Forty-two percent of the simulations resulted in predicted lesion emergence (i.e., the date of the actual symptoms onset overlaps, for at least one day, the window predicted by the model), while 52% resulted in prediction of no lesion emergence because the environmental conditions were not favorable for the completion of the process (Table 1). The remaining 6% were incorrect, with 29 false positive predictions and 1 false negative prediction. Lesion emergence was correctly predicted in 100% and 96% of the vineyards, in 2008 and 2009, respectively.

All observed lesion emergence were correctly predicted by the model, except for one, giving a true positive proportion (TPP) equal to 0.996 (Table 1), which represents the sensitivity of the model, i.e., the probability of a correct prediction of lesion emergence (17). Out of 313 simulations, 284 that predicted no infection were correct because no lesions were observed, giving a true negative proportion (TNP) of 0.907. This is the so called specificity of the model, which is the probability that the absence of infection is predicted correctly (17). Only one infection occurred in the vineyard without being predicted by the model; the false positive proportion (FNP) was 0.004 (Table 1). Finally, 29 out of 313 simulations predicted lesion emergence that did not result in observed lesions, giving a false negative proportion (FPP) of 0.093. Overall model accuracy was 0.945.

The posterior probability that there was an infection when the infection was predicted was $P(O+,P+) = 0.888$, and the probability that there was no infection when the infection was not predicted by the model was $P(O-,P-) = 0.996$, while the prior probabilities for infection and no infection were $P(O+) = 232 / 545 = 0.43$ and $P(O-) = 313 / 545 = 0.57$, respectively. The posterior probability that there was no infection when the model predicted an infection was $P(O-,P+) = 0.112$, and the posterior probability that there was infection when infection was not predicted by the model was $P(O+,P-) = 0.004$.

A representative example of the model simulations in 2008 is presented in Figure 7. The model predicted 14 oospore germination cohorts when the simulation was run with data from a weather station located in L'Acadie (45.29° N, -73.35°E). Weather data from this site were used for model validation in four different vineyards near this area as well. The weather station registered 14 rain events (total of 71.6 mm of rainfall) between 11 April and 28 May, which triggered the germination of 14 oospore cohorts (Fig. 6). The model predicted that six cohorts would produce lesions on four days with four different time-period for lesion emergence: 1 June (1 cohort) with predicted lesion emergence between 7 and 10 June; 2 June (2 cohorts) with predicted lesion emergence between 8 and 10 June; 3 June (1 cohort) with predicted onset between 8 and 11 June, 4 June (2 cohorts) with predicted onset between 11 and 13 June. The first symptoms of *P. viticola* in this area were observed in three vineyards, which were scouted at least every three days, on 9, 10, and 11 June. For these vineyards, the first three lesion emergences predicted by the model were correct. For the last location, the first three predicted lesion emergence were false positives, but the lesion emergence predicted on 4 June resulted in lesions observed in this vineyard on 13 June.

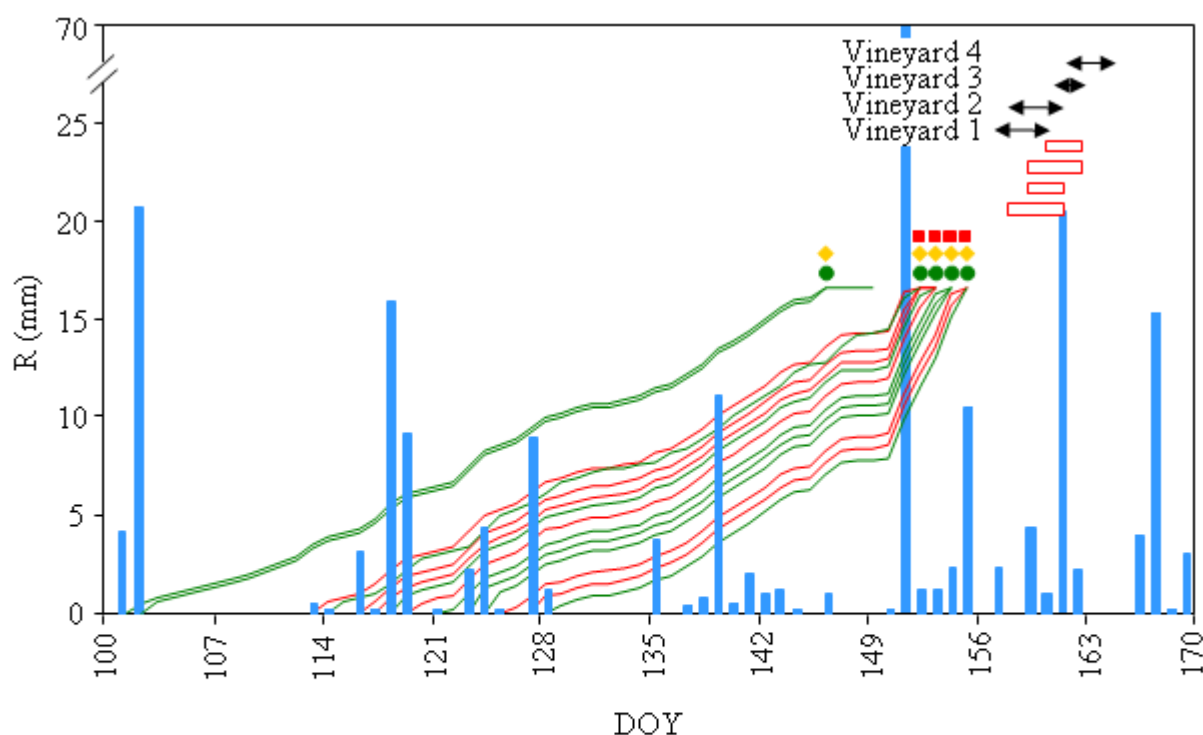


Fig. 6. Examples showing model simulations in the vineyards of L'Acadie (Montreal) in 2008. The blue bars represent rainfall, the lines (— and —) show the germination course of oospore cohorts which did or did not cause infection, respectively. Other symbols represent: ● zoospore release; ◆ zoospore dispersal; ■ infection; □ and ↔ estimated and actual periods of downy mildew onset, respectively.

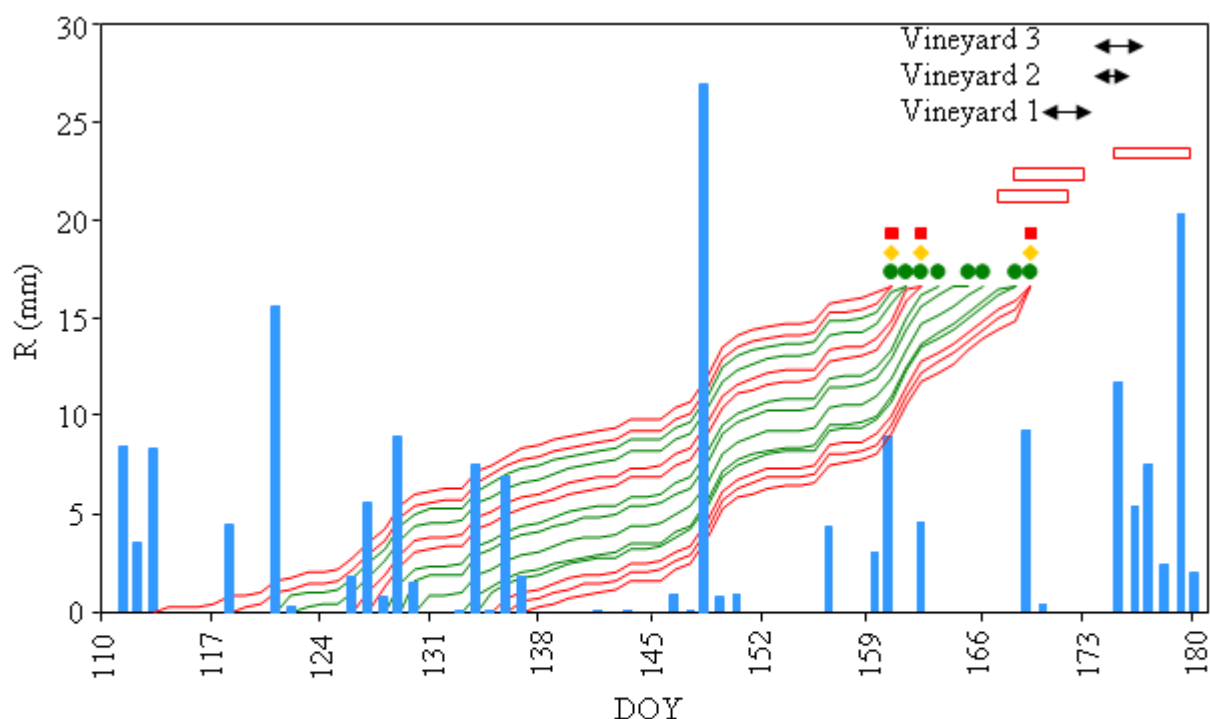


Fig. 7. Examples showing model simulations in the vineyards of Île d'Orléans (Quebec City) 2009. The blue bars represent rainfall, the lines (— and —) show the germination course of oospore cohorts which did or did not cause infection, respectively. Other symbols represent: ● zoospore release; ◆ zoospore dispersal; ■ infection; □ and ↔ estimated and actual periods of downy mildew onset, respectively.

In the northern part of province of Quebec, the model predicted 14 oospore germination events when operated with weather data from L'Île d'Orléans (46.96°N, -70.95°E) in 2009 (Fig. 7). The model predicted that seven of these oospore cohorts would result in infection and lesion emergence in three periods: 10 June (2 cohorts); 12 June (2 cohorts); and 19 June (3 cohorts) (Fig. 7). For the first four cohorts, the model predicted lesion emergence between 17 and 22 June and downy mildew was detected in a vineyard on 23 June after the last negative scouting on 19 June (Vineyard 1 in Fig. 7). So the model simulations for these 2 oospore cohorts were considered correct for the Vineyard 1, but as false positive predictions for the Vineyards 2 and 3. In these two vineyards downy mildew primary lesions were detected on 25 and 26 June, respectively, in agreement with the predicted lesion emergence on 19 June (Fig. 7).

Only one model prediction out of 285 that predicted no infection was incorrect (false negative prediction) because downy mildew symptoms appeared in the vineyard before the predicted period (Table 1). This particular event occurred in 2009 at a vineyard in the area of Dunham (45.13°N, -72.81°E). Going back from the incubation period from the date when the first symptom was observed, there was only one possible rainy event (12 June at 2 p.m.) that triggered the observed infection (Fig. 8). For this rainfall, the model simulated that oospores produced sporangia at 1 p.m., that favorable condition of temperature and relative humidity allowed zoospores to be released immediately, and that the following rain dispersed zoospores to leaves. Immediately after the rain, weather station recorded a drop in RH and an increase of 6°C in temperature that probably prevented wetness to be formed on leaves, which is a critical factor for *P. viticola* infection (Fig. 8). Therefore, the model predicted that the infection process was stopped. One can speculate that, even if weather data were representative of the area, particular microclimatic conditions favored a longer wet period and consequently infection established

in that specific circumstance. Nevertheless, the model provided useful information from the practical point of view of the disease management: it simulated the infection 14 days before the most probable actual one so, even though it missed the actual infection, an hypothetical spray program would have been operating anyway in that vineyard.

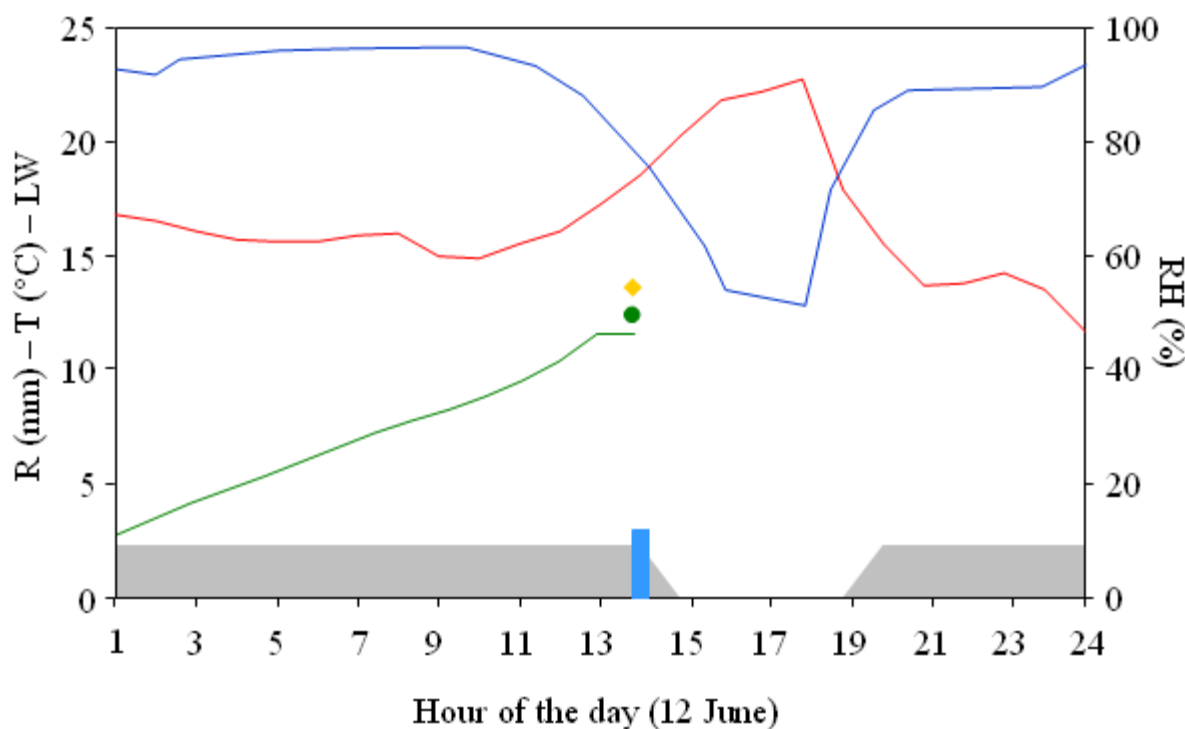


Fig. 8. Hourly weather conditions registered at Dunham, 12 June 2009, and corresponding model simulations. The green line (—) shows the final part of the germination course of the oospore cohort; ● zoospore release; ◆ zoospore dispersal; ■ presence of wetness; — temperature; — relative humidity; blue bar shows rainfall.

The three predictors calculated different threshold for timing the first fungicide application in the different locations (Fig. 9). The first infection simulated by the Kennelly rule ranged between 9 days before and 14 days after the actual infection: 32.6% of the first warnings by this rule were useless, occurring well more than four days after the actual infection (i.e., a period extending too long after infection to permit control by commonly used fungicides). The 3-10 rule increased the accuracy by identifying the first infection and the timing for first fungicide spray ranged from 8 days before to 6 days after the actual infection (Fig. 9). Even though 16.3% of the warnings produced were of no use, being either more than ten days before the actual infection (i.e., an hypothetical sprays would be wasted; 7.0% of the cases) or more than four days after the actual infection (9.3%). The UCSC model produced a further improvement because the first simulated infection was one day before the actual infection, in average, ranging between 6 days before and 3 days after (Fig. 9). Moreover, only one infection was predicted before the action threshold of ten days (one case in 2009) and the maximum delay registered was 3 days (only two cases, both in 2008).

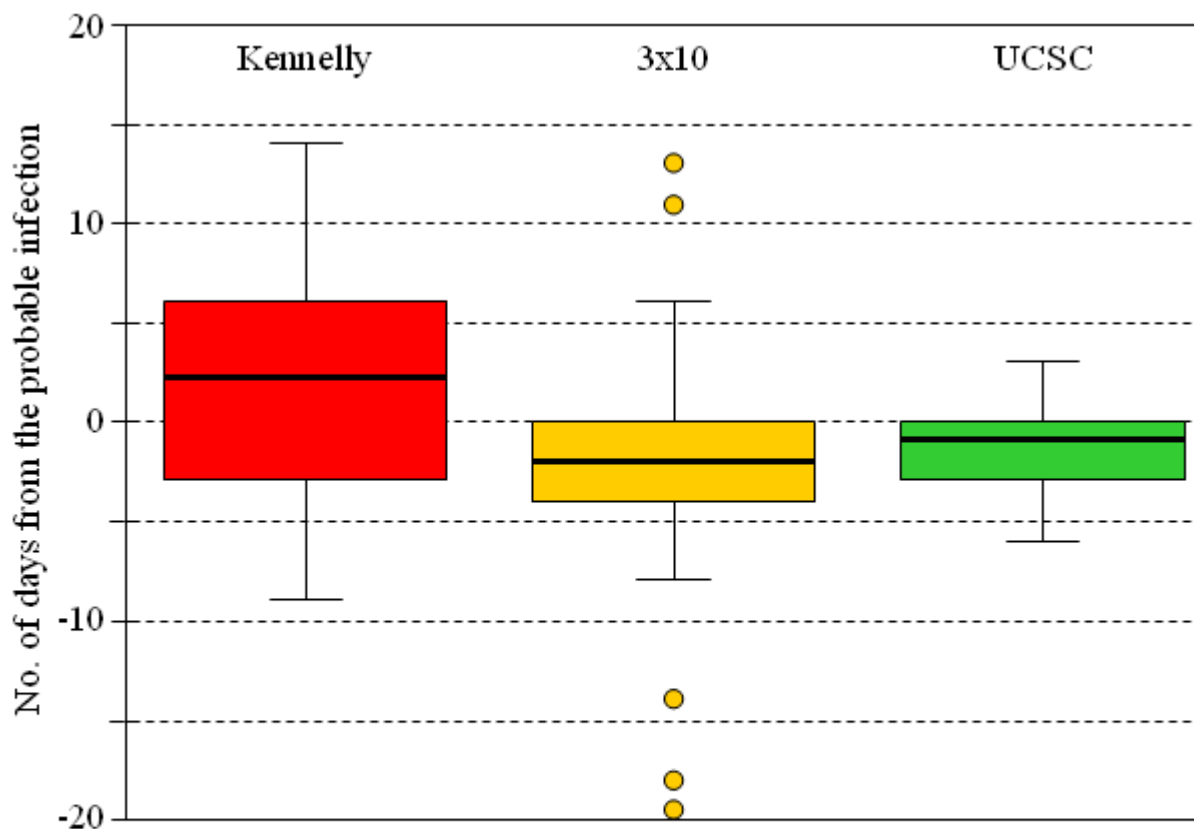


Fig. 9. Box-plot of the first seasonal infection of *Plasmopara viticola* calculated according the 3-10 rule (yellow), the Kennelly rule (red), and the UCSC model (green). For each box-plot is shown the median (—); dots are outliers.

Downy Mildew Management

Abundant rains during spring and summer of 2008 and 2009 resulted in severe epidemics of grape downy mildew in the province of Quebec. As a consequence, downy mildew is now well established in several vineyards in eastern Canada. Proper timing of fungicide applications will thus be crucial for adequate downy mildew management. The current strategy for management of grape downy mildew consists of initiating a calendar-based spray program generally in late spring or early summer using protective fungicide such as metiram, captan, or folpet (1). On highly susceptible cultivars in vineyard with significant infections of downy mildew in the previous year or when weather conditions are highly favorable, protective fungicides are replaced by curative fungicides such as metalaxyl-m or kresoxim-methyl (1). It is generally thought that the critical growth stages for downy mildew management are (on the Eichhorn-Lorenz scale): stage 9 (two to three leaves unfolded); just before bloom at stage 17 (inflorescence fully developed, flowers separating); and just after bloom at stage 27 (fruit set, young fruits beginning to swell). After these critical periods, fungicides are applied at 10- to 14-day intervals only if the weather is rainy and temperatures are favorable for infection. However, a number of fungicide sprays are applied as insurance against the highly erratic appearance of the disease and of the damages it causes. Several studies on the epidemiology of grapevine downy mildew have shown that the rate of development of the different phases of *P. viticola* life cycle is influenced by the weather conditions (14,18,20,23,25,26). Management of downy mildew should thus be based on disease risk rather than on host phenology. Infection predictions obtained operating with the empirical rules confirmed that a phenological threshold, such as the stage 12 on the Eichhorn-Lorenz scale or the

shoots length of 10 cm, is too risky for the growers, or may result in unneeded sprays. The former lead to an underestimation of the *P. viticola* primary infections (i.e., too late fungicide applications) while the latter is unstable and results are not enough reliable for scheduling disease control (i.e., 7% of predictions are more than 10 days before the actual infection). Even though the 3-10 rule may not result in untreated infections, it may lead to over use of fungicides early in the season. Furthermore, warning programs for management of grape downy mildew that ignore overwintering of the oospores and their maturation in the spring generally resulted in useless applications (16).

For almost a century, it was admitted that downy mildew epidemics are initiated by a small numbers of primary infections which occur early in the spring. After a rapid depletion of the stock of oospores, generally in June in eastern Canada, secondary infections drive the epidemic until leaf fall in autumn. This conviction stimulated the development of management strategies aiming at controlling the polycyclic phase of the disease caused by secondary inoculum. Recent work however, by Gobbin et al. (8,9) showed that in fact primary infections play an important role in downy mildew epidemic. In practice this mean that identifying and controlling the emergence of the primary lesions is crucial for downy mildew management. Hence, the practical objective of this study was to predict the emergence of the first lesions to eventually use this information to optimize timing of the first fungicide application. Based on the results of this validation study we concluded that the oospore simulation model, even though developed under weather conditions different than in eastern Canada, was reliable. However care must be taken in interpreting the results of this study because the predictions of lesion emergence were compared with field observations taken twice weekly, hence the lesions could has emerged up to 4 days earlier than what was reported from vineyard samplings. It will thus be important to further validate this model as an aid to manage grape downy mildew.

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